

41525 FEM – Day 4

Topology Optimization for Truss Structures

Ole Sigmund and Casper S. Andreasen

Department of Civil and Mechanical Engineering, Solid Mechanics

Preparatory reading:
Chapter 4 in Course Notes

Day 3 comments

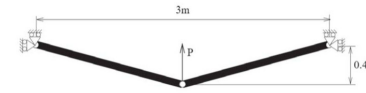
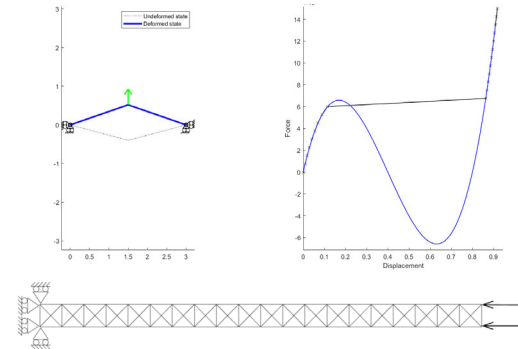


Figure 3.1: The 2-bar VonMises truss.



Recommendations:

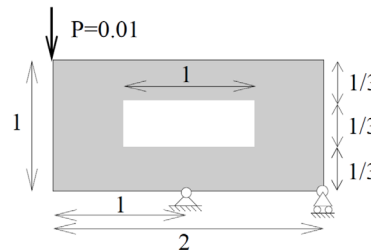
- Make backups
- Clean codes
- Follow pseudo codes
- Keep general
- Don't store everything
- Don't rename

Competition 1

Here is a competition: Who can make the stiffest truss structure? This corresponds to the truss structure with the smallest displacement (in the direction of the load) where the load is applied! ¹

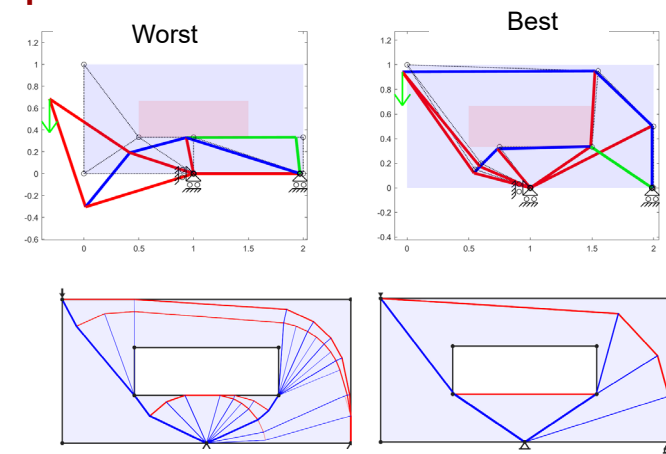
Rules: The truss structure should fit into the grey domain shown below and with no bars crossing the central void area (centered white region). Position and size of the vertical load is fixed (load is applied at the upper left corner). The center of the lower edge is supported in both x and y direction and the lower right corner only in the y-direction. The bar areas and Young's moduli are fixed to $A = 1$ and $E = 1$, respectively. The maximum total length of bar elements is 9 and you may use as many nodes and elements as you want. The stiffness matrix has to be positive definite.

Create an input file called "group#.m" (where # is your group number) and upload it to Learn, September 7th (by midnight). The teachers will evaluate the proposals and announce the prize-winner in the next lecture!

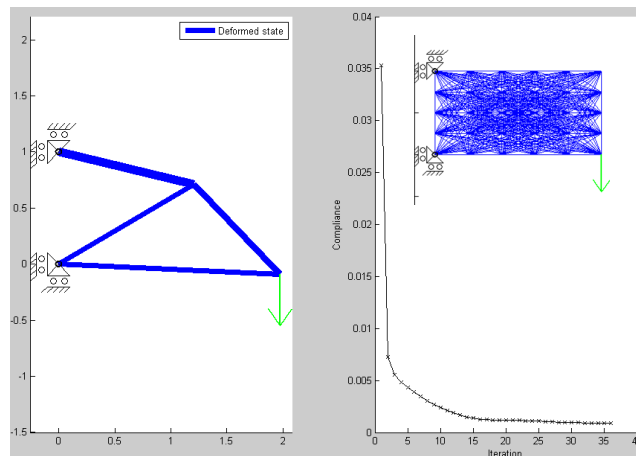
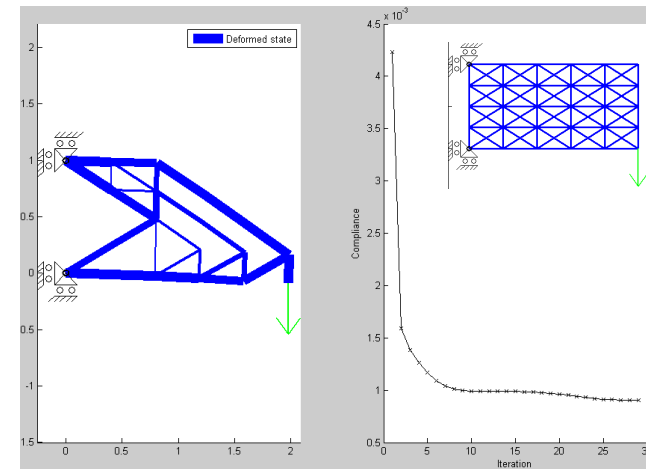
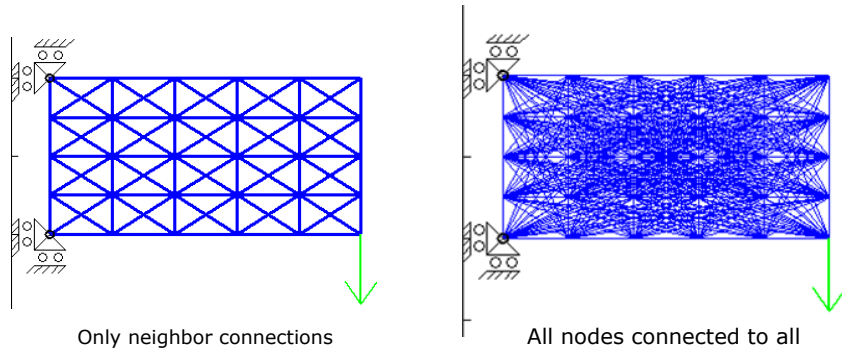


¹The submission (not the ranking!) is part of the evaluation of the first report in the course!

Competition results



Produced with <https://www.layopt.com/truss/>



combinations for picking M out of N bars:

$$\frac{N!}{(N - M)!M!}$$

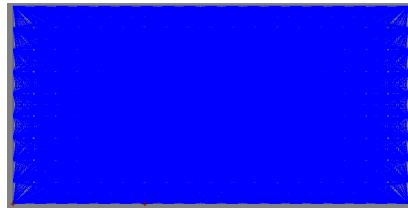
$$N=10, M=5 \Rightarrow 252$$

$$N=20, M=10 \Rightarrow 185.000$$

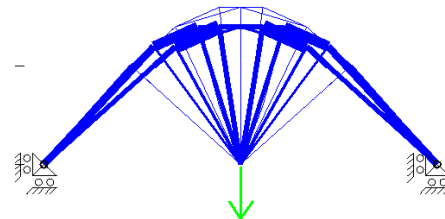
$$N=40, M=20 \Rightarrow 1.4 \cdot 10^9$$

$$N=100, M=50 \Rightarrow 10^{29}$$

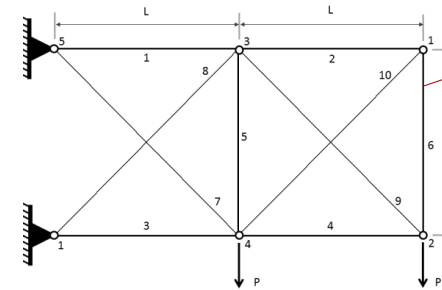
$$N=435, M=200 \Rightarrow 10^{152}$$



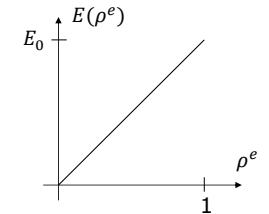
FlExtrac: 19x11 nodes, 21736 elements



Vol=8.00, C=8.5·10⁻⁵

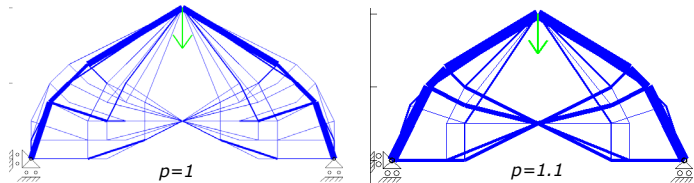


$$E^e = E(\rho^e) = (\rho^e) E_0$$

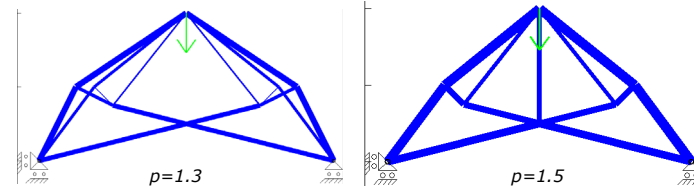


$$0 < \rho_{min} \leq \rho^e \leq 1$$

$$[k^e] = (\rho^e) [k_0^e]$$



17 x 9 (NE=10417)



$p \in [1, 1.5]$

Objective function :
 ↓ Maximize stiffness
 ↓ Minimize displacement
 ↓ Minimize compliance
 $C = \{D\}^T \{P\} = \{D\}^T [K] \{D\} = \sum_e \{d^e\}^T [k^e] \{d^e\}$

Volume Constraint : $V = \sum_e \rho^e v^e = \{\rho\}^T \{v\} \leq V^*$

Variables $\{\rho\}$: $\{0\} < \{\rho_{min}\} \leq \{\rho\} \leq \{1\}$

Stiffness – ρ relation : $E(\rho^e) = (\rho^e)^p E_0$ or $[k^e] = (\rho^e)^p [k_0^e]$

Stiffness matrix
for solid bar

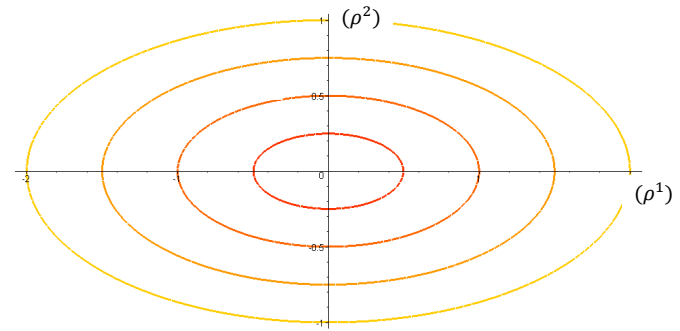
$$\begin{aligned} \min : & C = \{D\}^T [K] \{D\} \\ \text{s.t.} : & g = \{\rho\}^T \{v\} - V^* \leq 0 \\ & : \{0\} < \{\rho_{min}\} \leq \{\rho\} \leq \{1\} \\ & : [K(\{\rho\})] \{D\} = \{P\} \end{aligned}$$

$$[K(\{\rho\})] = \sum_e [k^e(\rho^e)] = \sum_e (\rho^e)^p [k_0^e]$$

Optimization crash course

Unconstrained optimization problem

$$\min_{\rho^1, \rho^2} f = (\rho^1)^2 + 4(\rho^2)^2$$



Optimality Criterion: unconstrained optimization problem

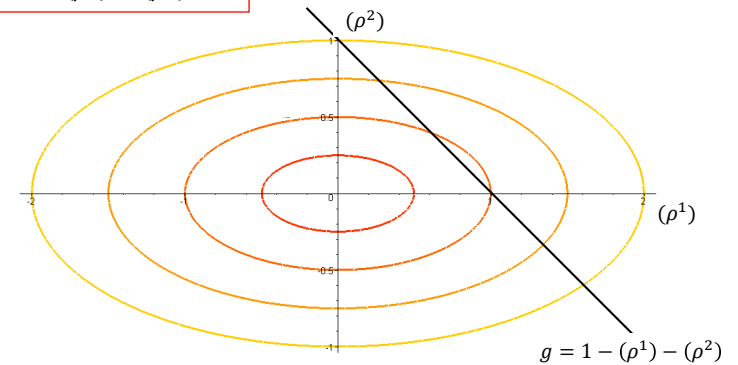
$$\nabla f = \{0\}$$

where: f Objective function

$$\text{Definitions: } \nabla f \equiv \begin{pmatrix} \frac{\partial f}{\partial \rho^1} \\ \frac{\partial f}{\partial \rho^2} \\ \vdots \\ \frac{\partial f}{\partial \rho^{NE}} \end{pmatrix}, \quad f' \equiv \frac{\partial f}{\partial \rho^e}$$

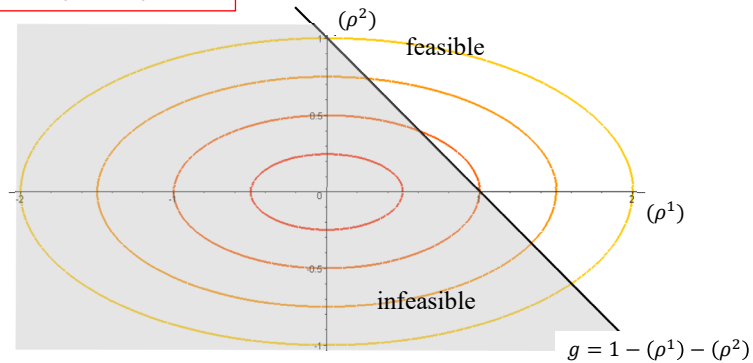
Constrained optimization problem

$$\begin{aligned} \min_{\rho^1, \rho^2} f &= (\rho^1)^2 + 4(\rho^2)^2 \\ \text{s.t.} \quad g &= 1 - (\rho^1) - (\rho^2) \leq 0 \end{aligned}$$



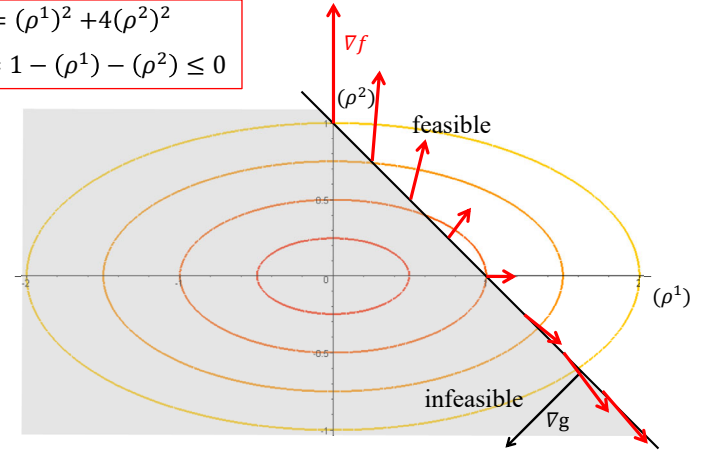
Constrained optimization problem

$$\begin{aligned} \min_{\rho^1, \rho^2} \quad & f = (\rho^1)^2 + 4(\rho^2)^2 \\ \text{s.t.} \quad & g = 1 - (\rho^1) - (\rho^2) \leq 0 \end{aligned}$$



Constrained optimization problem

$$\begin{aligned} \min_{\rho^1, \rho^2} \quad & f = (\rho^1)^2 + 4(\rho^2)^2 \\ \text{s.t.} \quad & g = 1 - (\rho^1) - (\rho^2) \leq 0 \end{aligned}$$



Optimality Criterion: constrained optimization problem

$$\nabla f + \lambda \nabla g = \{0\}$$

where:

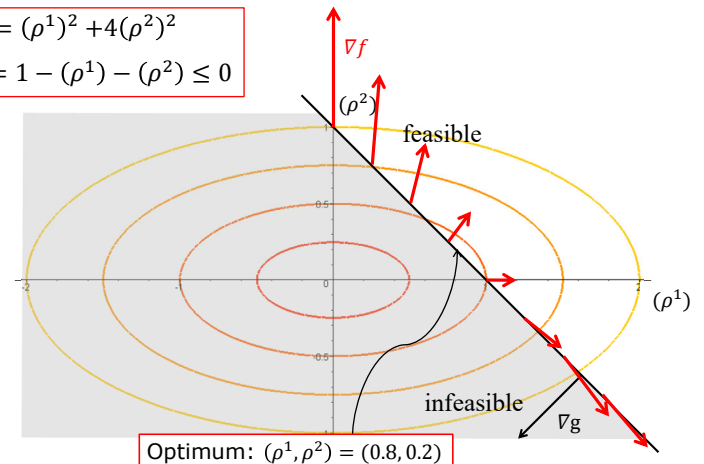
- f Objective function
- g Constraint function ($g \leq 0$)
- λ Lagrange multiplier

Definitions:

$$\nabla f \equiv \begin{Bmatrix} \frac{\partial f}{\partial \rho^1} \\ \frac{\partial f}{\partial \rho^2} \\ \vdots \\ \frac{\partial f}{\partial \rho^{NE}} \end{Bmatrix}, \quad f' \equiv \frac{\partial f}{\partial \rho^e}$$

Constrained optimization problem - solution

$$\begin{aligned} \min_{\rho^1, \rho^2} \quad & f = (\rho^1)^2 + 4(\rho^2)^2 \\ \text{s.t.} \quad & g = 1 - (\rho^1) - (\rho^2) \leq 0 \end{aligned}$$



Optimum: $(\rho^1, \rho^2) = (0.8, 0.2)$

Formulation of optimization problem

Objective function :
 ↓ Maximize stiffness
 ↓ Minimize displacement
 ↓ Minimize compliance
 ↓ $C = \{D\}^T \{P\} = \{D\}^T [K] \{D\} = \sum_e \{d^e\}^T [k^e] \{d^e\}$

Volume Constraint : $V = \sum_e \rho^e v^e = \{\rho\}^T \{v\} \leq V^*$

Variables $\{\rho\}$: $\{0\} < \{\rho_{min}\} \leq \{\rho\} \leq \{1\}$

Stiffness – ρ relation : $E(\rho^e) = (\rho^e)^p E_0$ or $[k^e] = (\rho^e)^p [k_0^e]$

Stiffness matrix
for solid bar

$$\begin{aligned} \min_{\{\rho\}} : & C = \{D\}^T [K] \{D\} \\ \text{s.t.} : & g = \{\rho\}^T \{v\} - V^* \leq 0 \\ & \{0\} < \{\rho_{min}\} \leq \{\rho\} \leq \{1\} \\ & [K(\{\rho\})] \{D\} = \{P\} \end{aligned}$$

$$[K(\{\rho\})] = \sum_e [k^e(\rho^e)] = \sum_e (\rho^e)^p [k_0^e]$$

Sensitivity analysis (constraint function)

Constraint function:

$$g = \{\rho\}^T \{v\} - V^* = \sum_e \rho^e v^e - V^* \leq 0$$

Gradient:

$$g' \equiv \frac{\partial g}{\partial \rho^e} = v^e \Rightarrow \nabla g = \{v\}$$

Sensitivity analysis (objective function) I

Objective function:

$$f = \{D\}^T [K] \{D\} = \sum_e \{d^e\}^T [k^e] \{d^e\} = \sum_e (\rho^e)^p \{d^e\}^T [k_0^e] \{d^e\}$$

1) Finite Difference (FD) approach:

$$\frac{\partial f}{\partial \rho^e} \approx \frac{\Delta f}{\Delta \rho^e} = \frac{f(\{\rho\} + \epsilon \{e\}_e) - f(\{\rho\})}{\epsilon}$$

where $\{e\}_e = \{0, 0, \dots, 1, \dots, 0\}^T$

NB! FD approach requires NE extra FE analyses ®

Sensitivity analysis (objective function) II

Objective function:

$$f = \{D\}^T [K] \{D\} = \sum_e \{d^e\}^T [k^e] \{d^e\} = \sum_e (\rho^e)^p \{d^e\}^T [k_0^e] \{d^e\}$$

2) Adjoint approach (using compact form $f = D^T K D$)

$$f' = D'^T K D + D^T K' D + D^T K D' = D^T K' D + 2 D^T K D' \quad (*)$$

"Trick" - differentiate FE equation ($K D = P$):

$$K' D + K D' = 0 \Rightarrow K D' = -K' D \quad (\text{assuming } P \text{ independent on design}) (**)$$

Insert (**) in (*)

$$f' = D^T K' D - 2 D^T K' D = -D^T K' D$$

DTU Sensitivity analysis (objective function) III

Objective function:

$$f = \{D\}^T [K] \{D\} = \sum_e \{d^e\}^T [k^e] \{d^e\} = \sum_e (\rho^e)^p \{d^e\}^T [k_0^e] \{d^e\}$$

2) Adjoint approach (cont'd)

$$\begin{aligned} f' &\equiv \frac{\partial f}{\partial \rho^e} = -D^T K' D = -D^T \frac{\partial K}{\partial \rho^e} D = -\{d^e\}^T \frac{\partial [k^e]}{\partial \rho^e} \{d^e\} \\ &= -p(\rho^e)^{p-1} \{d^e\}^T [k_0^e] \{d^e\} \end{aligned}$$

Adjoint approach requires no extra FE analysis ☺

DTU Optimality Criterion

Constraint gradient: $\frac{\partial g}{\partial \rho^e} = v^e \quad (*)$

Objective gradient: $\frac{\partial f}{\partial \rho^e} = -p(\rho^e)^{p-1} \{d^e\}^T [k_0^e] \{d^e\} \quad (**)$

Optimality Criterion: $\nabla f + \lambda \nabla g = \{0\} \Leftrightarrow$

$$\frac{\partial f}{\partial \rho^e} + \lambda \frac{\partial g}{\partial \rho^e} = 0, \quad e = 1, \dots, NE \Leftrightarrow$$

$$\frac{-\frac{\partial f}{\partial \rho^e}}{\lambda \frac{\partial g}{\partial \rho^e}} = 1, \quad e = 1, \dots, NE \quad (***)$$

Insert (*) and (**) in (***):

$$\frac{p \{d^e\}^T [k^e] \{d^e\}}{\lambda v^e \rho^e} = 1, \quad e = 1, \dots, NE$$

DTU Physical meaning of optimality condition

Optimality condition:

(holds for "active elements")

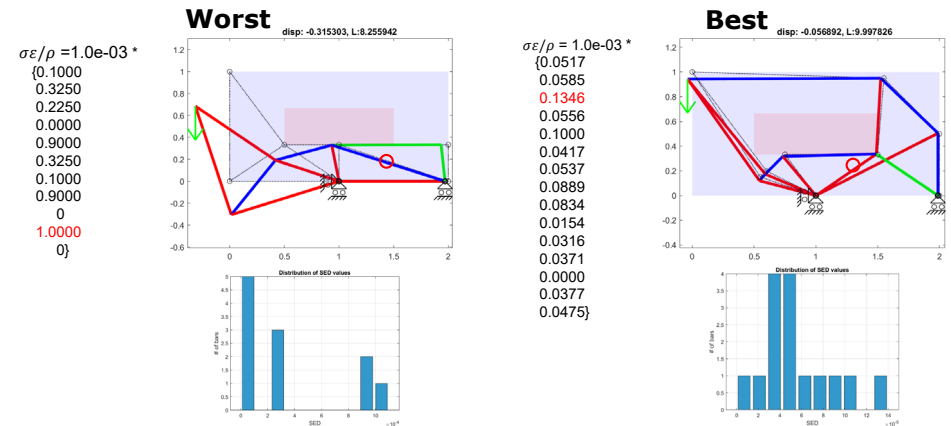
$$\frac{p}{\lambda} \frac{\{d^e\}^T [k^e] \{d^e\}}{v^e \rho^e} = 1, \quad e = 1, \dots, NE$$

i.e. strain energy pr. material volume must be constant

or

Strain Energy Density (SED) must be constant in all elements not meeting the lower or upper density bounds (i.e. for the active elements!)

DTU Competition results



Heuristic update rule

SED high => increase density

SED low => decrease density

$$\rho^e = \rho_{old}^e \left(\frac{-\frac{\partial f}{\partial \rho^e}}{\lambda \frac{\partial g}{\partial \rho^e}} \right)^\eta = \rho_{old}^e (B^e)^\eta, \text{ where } B^e = \frac{-\frac{\partial f}{\partial \rho^e}}{\lambda \frac{\partial g}{\partial \rho^e}} = \frac{p\{d^e\}^T [k^e] \{d^e\}}{\lambda v^e \rho^e}$$

$$\rho^e = \begin{cases} \rho_{min} & \text{if } \rho_{old}^e (B^e)^\eta \leq \rho_{min} \\ \rho_{old}^e (B^e)^\eta & \text{otherwise "active elements"} \\ 1 & \text{if } \rho_{old}^e (B^e)^\eta \geq 1 \end{cases}$$

Satisfying the volume constraint

$$\rho^e = \rho_{old}^e \left(\frac{-\frac{\partial f}{\partial \rho^e}}{\lambda \frac{\partial g}{\partial \rho^e}} \right)^\eta$$

$$g = g(\lambda) = g(\{\rho(\lambda)\}) = \{\rho(\lambda)\}^T \{v\} - V^* \leq 0$$

Bi-section scheme

Guess extremal values of λ (e.g. $\lambda_1 = 10^{-10}$ and $\lambda_2 = 10^{10}$)

While $(\lambda_2 - \lambda_1)/(\lambda_1 + \lambda_2) > \varepsilon$ (e.g. $\varepsilon = 10^{-5}$)

$$\lambda_{mid} = (\lambda_1 + \lambda_2)/2$$

if $g(\lambda_{mid}) > 0$

$$\lambda_1 = \lambda_{mid}$$

else

$$\lambda_2 = \lambda_{mid}$$

end

end while

Guess extremal values of λ (e.g. $\lambda_1 = 10^{-10}$ and $\lambda_2 = 10^{10}$)

While $(\lambda_2 - \lambda_1)/(\lambda_1 + \lambda_2) > \varepsilon$ (e.g. $\varepsilon = 10^{-5}$)

$$\lambda_{mid} = (\lambda_1 + \lambda_2)/2$$

if $g(\lambda_{mid}) > 0$

$$\lambda_1 = \lambda_{mid}$$

else

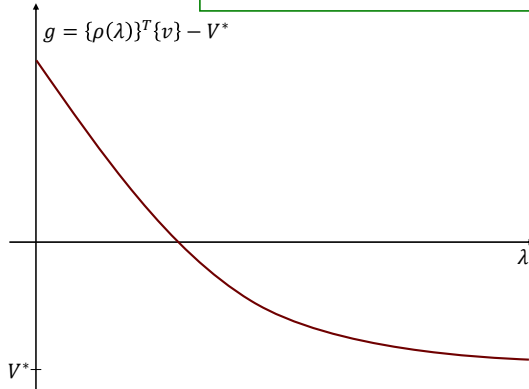
$$\lambda_2 = \lambda_{mid}$$

end

end while

$$\rho^e = \begin{cases} \rho_{min} & \text{if } \rho_{old}^e (B^e)^\eta \leq \rho_{min} \\ \rho_{old}^e (B^e)^\eta & \text{otherwise "active elements"} \\ 1 & \text{if } \rho_{old}^e (B^e)^\eta \geq 1 \end{cases} \quad \rho^e = \rho_{old}^e \left(\frac{-\frac{\partial f}{\partial \rho^e}}{\lambda \frac{\partial g}{\partial \rho^e}} \right)^\eta$$

$$g = \{\rho(\lambda)\}^T \{v\} - V^*$$



Topology optimization algorithm

Initialize $\{\rho\}$ and V^* (choose initial $\{\rho\}$ to satisfy V^*)

for iopt = 1, 2, ... max_iopt

Set $\{\rho_{old}\} = \{\rho\}$

Solve $[K]\{D\} = \{P\}$

Calculate $\frac{df}{d\rho^e}$ and $\frac{dg}{d\rho^e}$

Find $\{\rho\}$ by bi-section method (separated function)

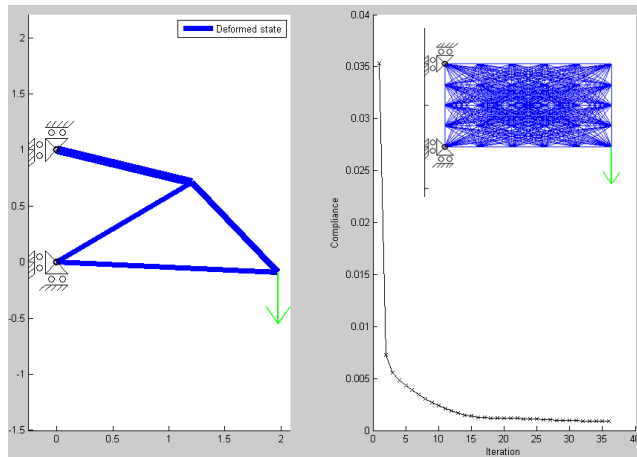
If $||\{\rho_{old}\} - \{\rho\}|| < \varepsilon ||\{\rho\}||$ break

Plot convergence ($f(\text{iopt})$)

Plot structure (e.g. with bar thicknesses proportional to relative element densities)

end

DTU Optimized structure for full connectivity



Exercise 4.1

Implement the Optimality Criteria approach for topology optimization of truss structures modelled by small displacements and strains.

Hint 1: Call a function "bisect", that uses the bi-section method to return the value of the design variable vector $\{\rho\}$ as a function of $\{\rho_{old}\}$, V^* , $\{\frac{df}{d\rho}\}$ and $\{\frac{dg}{d\rho}\}$, i.e.

$$[\{\rho\}] = \text{bisect}(\{\rho_{old}\}, V^*, \{\frac{df}{d\rho}\}, \{\frac{dg}{d\rho}\})$$

Program this function in a separate file named 'bisect.m' for example. In Matlab, functions begin with the line
function [outdata, ..] = bisect(indata, ..)

Test the program on the example from Exercise 1.1 (Fig. 1.3 but add the two missing diagonal bars) with $\rho_{min} = 10^{-6}$, $V^* = 10$ and $P = 0.01$. Plot the compliance as a function of the iteration number. Check that the strain energy density (e.g. $\sigma \epsilon / \rho$) is constant for each active (i.e. not governed by the lower or upper density bound) element. (NB! remark that the element stress now should be computed as $\sigma = E \epsilon = (\rho^*)^p E_0 \epsilon$).

i.e. not ρ_{min} or 1 !!

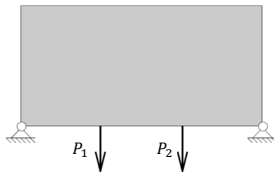
Exercise 4.2

Study the effect of varying the damping parameter η ? Which value should be chosen for fastest convergence? Use a bigger example than the simple 11-bar truss for this study.



*Exercise 4.3

Implement and solve multi-loadcase problems (objective is weighted sum of compliances for different load cases). What is the difference in applying the loads simultaneously or independently? Hint: minimize either $f_1 = \{D_{1+2}\}^T (\{P_1\} + \{P_2\})$ where $[K]\{D_{1+2}\} = \{P_1\} + \{P_2\}$ or $f_2 = \{D_1\}^T \{P_1\} + \{D_2\}^T \{P_2\}$ where $[K]\{D_1\} = \{P_1\}$ and $[K]\{D_2\} = \{P_2\}$. In words: in the first case the two loads are applied simultaneously, in the second case the two loads are applied in two different load cases.

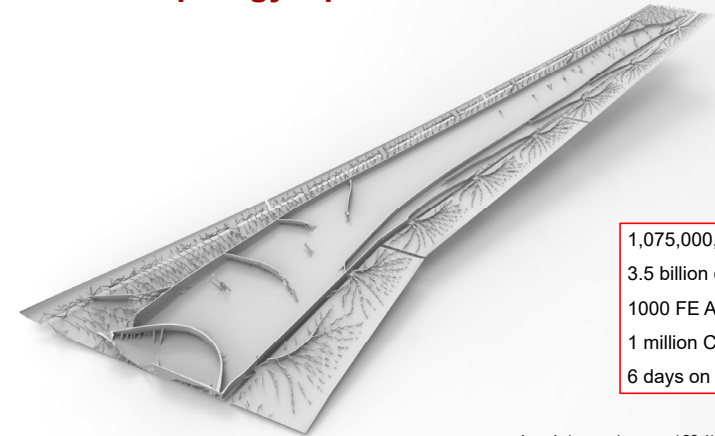


*Exercise 4.4

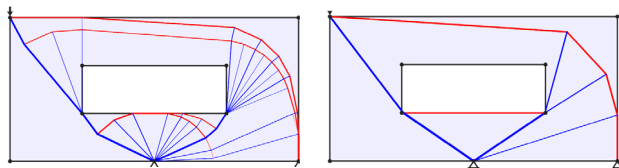
Implement everything in three dimensions and demonstrate by an example.



Advanced Topology Optimization Lectures on DAY10/DAY11



1,075,000,000 elements
3.5 billion dofs
1000 FE Analyses
1 million CPU hours
6 days on 6000 processors



Produced with <https://www.layopt.com/truss/>

- This first assignment consists of four subassignments in total (one per course day).
- Each subassignment must be answered on one separate page including figures and discussions. Also include subassignment numbers in answers, e.g. 3.1, 3.2, 3.e.
- Apart from the 4 pages discussed above, a cover page must include the following:
 - Group number and code.
 - Student's names and student's numbers.
 - A statement on who is responsible for what parts of the assignment and codes¹.
- One (in total) additional and separate page may be added containing answers to "exercises related to the subassignments. Recall that one can still achieve top grades without solving any "exercises".
- Use minimum 12 pt, single line spacing, 2cm margin left and right, 3cm margin top and bottom on your A4 paper. (NB! This document is formatted according to these rules)
- The report must be uploaded to DTU Learn as *one separate pdf-file* before the deadline (Tuesday September 30th at 10pm).
- One separate pdf-file containing all source codes for one representative Matlab/Python code pr. subassignment must be uploaded to DTU Learn as well.
- Failure to adhere to above rules will count negatively in the evaluation.
- Please follow the DTU guidelines for the use of generative AI in project reports as found here: <https://student.dtu.dk/en/exam/exam-cheating/dtu-code-of-honour> and <https://student.dtu.dk/en/exam/exam-cheating>.
- Remember that AI cannot be a co-author of your assignment!
- While AI tools may be used to support tasks such as coding, data processing, or drafting text, the primary responsibility for the submitted work must be able to explain, justify, defend and explain the codes, methods, analyses, and conclusions presented in the report.

Recommendations:

- Use the feedback sheet (see the sample outline next page) as work guidance and check-list.
- The one page answers to each subassignment may contain plots, graphs, equations and text.
- All results must be accompanied by short and precise explanations, discussions and critical interpretations. Note that these discussions and interpretations also count in the evaluation.
- Create simple, clear and readable illustrations and graphs.

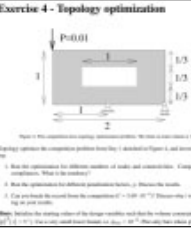
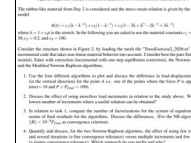
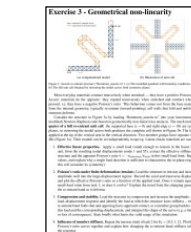
¹From the DTU rules: "It must be clearly specified for which sections each student has the (main) responsibility. A group project is not deemed to be individualized if the students merely state that they have contributed equally to all sections of the report or the like. If a group project does not comply with these requirements for individualization or other formal requirements, the report may be rejected and no assessment given."

Group code:

Feedback sheet

	uk	MA	Unacceptable just acceptable	Average	Good	Excellent	Remarks
Report specific issues							
Layout and format	x						
Maths codes	x						
Cover page							Use just 1 instead of 3
Page limit	x						
Competition 1	x						
Clarity and precision language					x		
Revisiting of graphs							fonts in figs 3-5 too small
Code organization			x				
Subsessment 2							
Ex. 1					x		
Ex. 2					x		
Ex. 3					x		
Ex. 4			x				Vertical bars compressed to 300MP
Ex. 5							please include sketch or more details
Ex. 6			x				discuss a bit more
Subsessment 3							
Ex. 1					x		indicate datapoints!
Ex. 2					x		
Ex. 3					x		
Ex. 4					x		
Subsessment 3							
Ex. 1					x		
Ex. 2					x		
Ex. 3					x		
Ex. 4					x		Actually Euler VI Code correct - perfect
Subsessment 4							
Ex. 1			x				convergence issues
Ex. 2							What is C7 should plot with lineplot
Ex. 3			x				

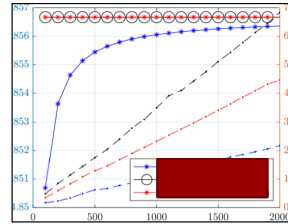
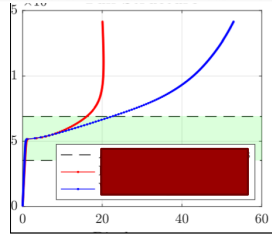
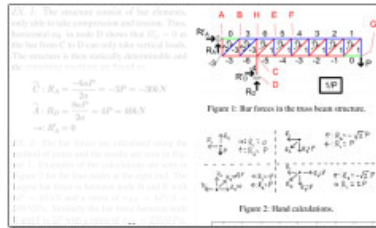
Nice report. Just a few small issues



Full Assignment 1 released on Learn at 3pm today



Examples from prior reports



Office hours before First Assignment

(Check Learn for latest updates / changes)

Person	Thursday morning (24/9)	Thursday afternoon (24/9)	Friday morning (25/9)	Friday afternoon (25/9)	Monday morning (28/9)	Monday afternoon (28/9)	Tuesday morning (29/9)	Tuesday afternoon (29/9)
Ole						14-15 (404/136)		
Casper				14-15 (404/010)				
Victor								14-15 (404-127)
Brandon		14-15 (B404/R117)						
Jonathan			10-11 (B404/R121)					
Junpeng							11-12 (404-126)	

NB! Please come well prepared!

Deadline Tuesday, September 29th, 10pm!!



Kahoot